

Sequential Quantum Instruments

Depletion, Regeneration, and Minimal Predictive State

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Abstract

Sequential measurement is analysed through the complete instrument object: an outcome map and its conditional successor state. A sharp alternating walk on one singlet is solved exactly, proving that the first local result destroys the reusable Bell resource while later conditional entropy remains positive. Repeated unread projective channels are classified by their fixed operator algebra, yielding the exact stationary witness value m/d and the sharp rank-over-dimension compatibility floor. A Poisson equation gives the finite-time passage potential and the exact criterion for sustained contextual witnesses; the Yu-Oh no-reset protocol and KCBS selective-replacement thresholds are derived. A processed four-outcome Bell instrument is then physically completed and iterated in a common basis. Its current mark is the unique minimal predictive state, with exact stationary, spectral, information, and record-cost values.

Keywords: quantum instrument; sequential measurement; Bell nonlocality; contextuality; predictive state; entropy rate

Scope and status: Analytic theorems retain their displayed protocol assumptions. Processed-matrix results are exact for the supplied matrices and positivity completion. No result is generalized to all measurement systems.

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1 Operational object

A measurement operation is represented by a quantum instrument, not by a probability distribution alone. The instrument produces both an outcome and a conditional successor state. The same distinction is required when a sequence is reused: outcome uncertainty, state memory, nonclassical certification, and physical record cost are separate quantities.

For a completely positive instrument $\{\mathcal{I}_a\}_a$,

$$p(a | \rho) = \text{Tr } \mathcal{I}_a(\rho), \quad \rho'_a = \frac{\mathcal{I}_a(\rho)}{p(a | \rho)}. \quad (1)$$

The unread channel is $\mathcal{T} = \sum_a \mathcal{I}_a$. All repeated-use calculations below preserve this complete object.

Part I Sharp sequential measurement of one singlet

2 Even-cycle protocol

Let $n \geq 4$ be even, $\delta = \pi/n$, $c = \cos \delta$, and $d = \cos(2\delta)$. Define

$$O(\theta) = \cos \theta \sigma_z + \sin \theta \sigma_x, \quad (2)$$

and start from the singlet

$$|\Psi^-\rangle = \frac{|01\rangle - |10\rangle}{\sqrt{2}}. \quad (3)$$

At global step k , measure $O(k\delta)$ on Alice for even k and Bob for odd k , recording $X_k \in \{-1, +1\}$. Adjacent measurements act on different subsystems and are compatible; every measurement is rank-one and projective.

Theorem 2.1 (Sharp-mark depletion). *Conditional on $X_0 = a$, the post-measurement joint state is*

$$\rho_{AB}^{(a)} = |a; 0\rangle\langle a; 0|_A \otimes |-a; 0\rangle\langle -a; 0|_B. \quad (4)$$

Its entanglement negativity is zero. Every later selected local projective outcome keeps the state product. The same pair cannot generate a later Bell violation unless a new entangling operation is applied.

Proof. The singlet is perfectly anticorrelated in every common basis: $(P_a(0) \otimes I)|\Psi^-\rangle \propto |a; 0\rangle_A \otimes |-a; 0\rangle_B$. This vector has Schmidt rank one. A local rank-one projector maps a product vector to another product vector or zero, so induction covers the remaining selected outcomes. The unread first measurement is a convex mixture of product states and is separable. $\square$

3 Exact entropy profile

The first result is uniform. Singlet anticorrelation and the local two-step update give

$$\Pr(X_1 = b | X_0 = a) = \frac{1 - abc}{2}, \quad (5)$$

$$\Pr(X_k = x | X_{<k}) = \Pr(X_k = x | X_{k-2} = y) = \frac{1 + xyd}{2}, \quad k \geq 2. \quad (6)$$

With binary entropy h_2 ,

Theorem 3.1 (Entropy profile).

$$H(X_0) = 1, \quad (7)$$

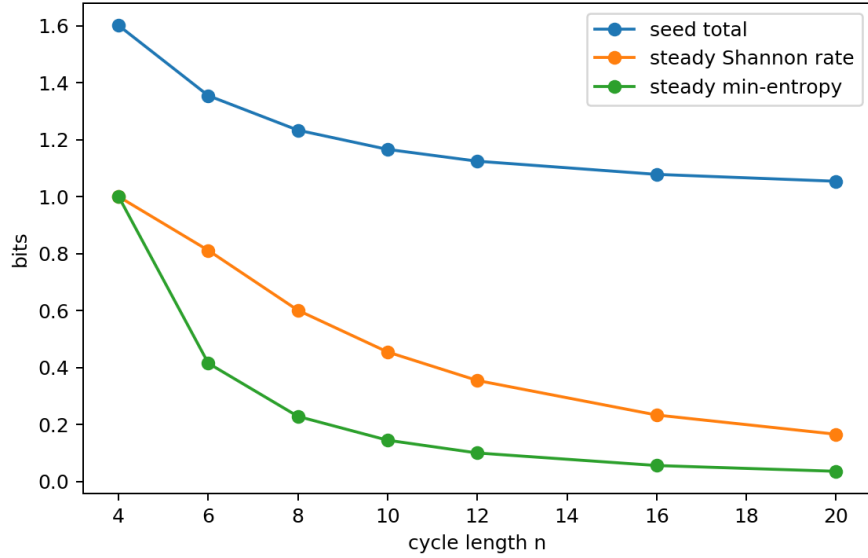
$$H(X_1 | X_0) = h_2\left(\sin^2 \frac{\pi}{2n}\right), \quad (8)$$

$$H(X_k | X_{<k}) = h_2\left(\cos^2 \frac{\pi}{n}\right), \quad k \geq 2. \quad (9)$$

The trusted-model guessing probability after the seed is $\cos^2(\pi/n)$ and the conditional min-entropy is $-\log_2 \cos^2(\pi/n) > 0$.

Table 1: Exact entropy values for representative even cycles.

n	$1 + H(X_1 X_0)$	steady Shannon rate	steady guessing probability
4	1.6009	1.0000	0.5000
6	1.3546	0.8113	0.7500
8	1.2333	0.6009	0.8536


Figure 1: Seed entropy, steady Shannon rate, and trusted-model min-entropy.

4 Geometric correlator depletion

Define $R_k = \mathbb{E}[X_k X_{k+1}]$. Since $\mathbb{E}[X_{k+2} | X_k] = dX_k$,

Theorem 4.1 (Geometric decay).

$$R_k = -\cos \frac{\pi}{n} \cos^k \frac{2\pi}{n}. \quad (10)$$

For $n = 4$, all correlators after R_0 vanish exactly. The signed sum over every later n -step block tends to zero.

A fresh-singlet experiment on every edge instead gives the chained magnitude $n \cos(\pi/n)$. These are different resource schedules, not two readings of the same sequential resource.

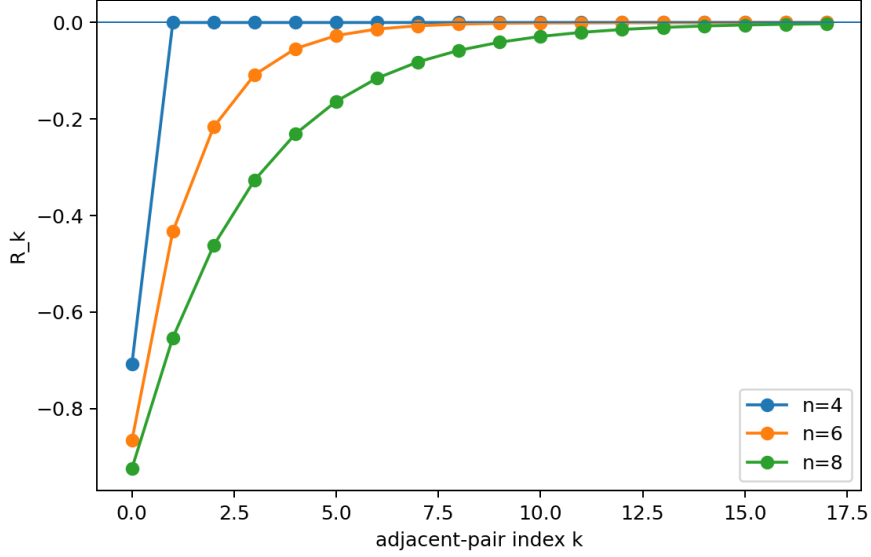


Figure 2: Exact adjacent-correlator decay for three even cycles.

Theorem 4.2 (Certificate separation). *Positive post-seed conditional entropy does not imply a post-seed Bell-device-independent certificate from the same walked pair.*

Proof. After the first result the conditional joint state is product. The remaining process has a local causal realization: sample the correlated seed pair, then update the two sides with independent local transition coins. The later statistics therefore do not require a nonlocal resource, while the Shannon and trusted-model min-entropies remain positive. $\square$

Part II Repeated contextual instruments

5 Unital self-idling

For a projector P , the unread Lueder channel

$$\mathcal{D}_P(X) = PXP + (I - P)X(I - P) \quad (11)$$

is an orthogonal projection in Hilbert-Schmidt operator space. For a periodic sequence $P_0, \dots, P_{m-1}$, let $\mathcal{T} = \mathcal{D}_{P_{m-1}} \cdots \mathcal{D}_{P_0}$.

Theorem 5.1 (Unital self-idling). *The fixed operator space is the intersection of the commutants of the P_j . If the family is irreducible, $\mathcal{T}^k(\rho) \rightarrow I/d$ for every initial state. An unweighted rank-one witness therefore idles at*

$$W_{\text{idle}} = m/d. \quad (12)$$

This explains the exact values $5/3$, $7/5$, and 2 in the audited walks. Comparing m/d with the noncontextual bound $\alpha(G)$ gives strict idling, facet idling, or state-independent violation.

Theorem 5.2 (Adjacent-orthogonality floor). *If $P_j P_{j+1} = 0$ and the stationary state is I/d , the asymptotic fraction of forced next steps obeys*

$$\tau \geq \frac{1}{m} \sum_j \frac{\text{rank } P_j}{d}. \quad (13)$$

For rank-one tests the sharp floor is $1/d$. The frequently observed value $1/3$ is the qutrit instance, not a universal constant.

6 Passage potential and stationary engine criterion

For randomized contexts c and score $g(c, x)$, define the averaged reward operator R and unread channel $\mathcal{T}$. If $\mathcal{T}$ is primitive with stationary state σ , put $q = \text{Tr}(\sigma R)$.

Theorem 6.1 (Poisson passage potential). *There is a Hermitian K , unique up to a scalar, satisfying*

$$K - \mathcal{T}^*(K) = R - qI. \quad (14)$$

For the mean evolution,

$$\sum_{t=0}^{N-1} \text{Tr}(\rho_t R) = Nq + \text{Tr}(\rho_0 K) - \text{Tr}(\rho_N K). \quad (15)$$

The long-run witness exceeds the noncontextual bound exactly when $q > \beta_{\text{NC}}$.

If $R = qI$, then $K = 0$ and the expected witness is q at every round for every state; no return to a preferred state is required.

7 Yu-Oh and KCBS calculations

For the 13-ray Yu-Oh construction,

$$L_{YO} = \frac{25}{3}I, \quad (16)$$

while the noncontextual bound is 8. With the displayed singleton/edge randomization, the quantum mean is $1/3$, the noncontextual mean is at most $8/25$, and the per-round gap is $1/75$.

For the five-ray KCBS family, replacement after a no outcome is optimal within the audited outcome-selective family. The exact thresholds are

$$r_{\text{no}}^* = 0.8396425434, \quad (17)$$

$$c_{\text{min}} = 0.5037855260, \quad (18)$$

$$r_{\text{uniform}}^* = 0.5115311845. \quad (19)$$

Replacement after yes outcomes alone reaches only $1.9098300563 < 2$.

Part III A complete four-outcome marked instrument

8 Physical completion and common basis

Let

$$E_a = \sum_r e_{ar} |v_{ar}\rangle\langle v_{ar}|, \quad \sigma_a = \sum_s q_{as} |w_{as}\rangle\langle w_{as}|. \quad (20)$$

The Kraus family

$$K_{asr} = \sqrt{e_{ar} q_{as}} |w_{as}\rangle\langle v_{ar}| \quad (21)$$

realizes $\mathcal{I}_a(\rho) = \text{Tr}(E_a \rho) \sigma_a$. The processed realization contains 60 nonzero Kraus operators. The complete classical-quantum Choi operator is

$$\mathbb{J} = \sum_a |a\rangle\langle a|_C \otimes \sigma_a \otimes E_a^\top. \quad (22)$$

One nonpositive tomography estimate is repaired by retaining Bell-basis populations and scaling coherences by the maximal positivity factor

$$\eta_* = 0.708418699729814. \quad (23)$$

Effects and conditional states must be transformed into one common basis before repeated-use multiplication. Omitting this transformation changes the transition law by Frobenius norm 0.541915971853 .

9 Corrected transition law and minimal predictive state

The corrected column-stochastic law is

$$C = \begin{pmatrix} 0.43696297 & 0.19810609 & 0.20644079 & 0.14060033 \\ 0.16017410 & 0.46165671 & 0.13951861 & 0.22362294 \\ 0.24443666 & 0.11892971 & 0.46061556 & 0.17659060 \\ 0.15842628 & 0.22130750 & 0.19342504 & 0.45918613 \end{pmatrix}. \quad (24)$$

Every pair of columns has positive total-variation distance, so no two marks can be merged in a first-order predictive state.

Theorem 9.1 (Minimal mark state). *The four current marks are the unique coarsest predictive states for all later repeated uses of the processed instrument.*

The stationary law and nontrivial eigenvalues are

$$\pi = (0.24327934, 0.24563854, 0.24991431, 0.26116781), \quad (25)$$

$$\text{spec}(C) = \{1, 0.35301358, 0.27498875, 0.19041902\}. \quad (26)$$

Writing $\Pi = \pi \mathbf{1}^\top$ and $R = C - \Pi$ gives $C^n = \Pi + R^n$.

10 Information loss and continuing entropy

For a uniform initial source label J ,

$$I(J; A_0) = 0.4720622220 \text{ bits}, \quad (27)$$

$$I(J; A_1) = 0.0409311197 \text{ bits}. \quad (28)$$

Only 8.6707% remains after one reuse. At stationarity,

$$H(A_t) = 1.9994586368 \text{ bits}, \quad (29)$$

$$h = H(A_{t+1} | A_t) = 1.8413629168 \text{ bits/use}, \quad (30)$$

$$I(A_t; A_{t+1}) = 0.1580957201 \text{ bits}. \quad (31)$$

The best next-mark probability is 0.4547437955, giving conditional min-entropy 1.1368741409 bits.

For surprisal, the centred Poisson potential is

$$k = (0.03023948, -0.02314357, 0.00046185, -0.00684276), \quad (32)$$

and the finite-time mean error is at most $0.06047896/N$ bits per use.

11 Record cost

With the previous mark available as side information, optimal cyclic compression and erasure obey

$$Q_{\min} \geq k_B T \ln 2 h = 1.2763355141 k_B T \quad (33)$$

per use. Avoiding reset requires memory growth at 1.8413629168 bits per use.

12 Scope and exact claim boundary

The singlet and contextuality theorems apply to their displayed protocols. The four-outcome results are exact for the processed public matrices and the displayed positive completion. No universal claim about all sequential measurements follows. Raw conditional entropy, trusted-model min-entropy, nonclassical witness value, source memory, and physical record cost remain separate ledger entries.

A Reproducibility table

Object	Check
Even-cycle depletion	product state after first result; negativity $1/2 \rightarrow 0$
Entropy	exact $n = 4, 6, 8$ values and positive post-seed min-entropy
Correlator	$R_k = -cd^k$ and geometric block extinction
Unital cycles	fixed commutant, m/d idling, rank/ d floor
Contextual regeneration	Yu-Oh operator identity and KCBS thresholds
Marked instrument	positivity repair, Kraus completeness, Choi positivity, basis conversion, stationarity, information decay, Poisson potential

B Full depletion and certificate proofs

Let $P_a(\theta) = (I + aO(\theta))/2$. The singlet satisfies perfect anticorrelation in every common basis:

$$(P_a(0) \otimes I)|\Psi^-\rangle \propto |a;0\rangle_A \otimes |-a;0\rangle_B. \quad (34)$$

This selected state has Schmidt rank one and zero negativity. The unread first result is the convex mixture of the two selected product states and is separable. Every later selected local rank-one projector preserves product form. Thus the first sharp record, not the second, consumes the same pair's future Bell resource.

Singlet anticorrelation gives

$$\Pr(X_1 = b \mid X_0 = a) = \frac{1 - ab \cos(\pi/n)}{2}, \quad (35)$$

and for $k \geq 2$ the most recent result on the same side is sufficient:

$$\Pr(X_k = x \mid X_{<k}) = \frac{1 + xX_{k-2} \cos(2\pi/n)}{2}. \quad (36)$$

Hence

$$H(X_0) = 1, \quad (37)$$

$$H(X_1 \mid X_0) = h_2\left(\sin^2 \frac{\pi}{2n}\right), \quad (38)$$

$$H(X_k \mid X_{<k}) = h_2\left(\cos^2 \frac{\pi}{n}\right), \quad k \geq 2. \quad (39)$$

The trusted-model guessing probability is $\cos^2(\pi/n)$ and the conditional min-entropy is its negative base-two logarithm.

The adjacent correlator obeys

$$R_{k+1} = \cos(2\pi/n)R_k, \quad R_0 = -\cos(\pi/n), \quad (40)$$

therefore

$$R_k = -\cos \frac{\pi}{n} \cos^k \frac{2\pi}{n}. \quad (41)$$

For $n = 4$, every correlator after the first vanishes exactly. A fresh-pair chained experiment instead supplies a new singlet at every edge and gives $n \cos(\pi/n)$. Those are different resource schedules, not two measurements of one conserved violation.

Table 3: *Exact entropy profile of the sharp even-cycle walk.*

n	seed $1 + H(X_1 X_0)$	steady rate	guess probability
4	1.6009	1.0000	0.5000
6	1.3546	0.8113	0.7500
8	1.2333	0.6009	0.8536

C Complete stationary witness derivation

For a sharp projector P , the unread channel

$$\mathcal{D}_P(X) = PXP + (I - P)X(I - P) \quad (42)$$

is the Hilbert–Schmidt orthogonal projection onto the commutant of P . A product of such channels fixes exactly the intersection of the commutants. If the projectors are irreducible, only scalar operators survive and every state converges to I/d . For m rank-one tests,

$$W_{\text{idle}} = \sum_{j=1}^m \text{Tr} \left(\frac{I}{d} P_j \right) = \frac{m}{d}. \quad (43)$$

If adjacent tests are orthogonal, every preceding yes result forces the next answer to be no. At stationarity the forced-step fraction satisfies

$$\tau \geq \frac{1}{m} \sum_j \frac{\text{rank } P_j}{d}, \quad (44)$$

which becomes $1/d$ for rank-one tests. The earlier apparent one-third constant is therefore the qutrit instance.

For a repeated instrument with averaged channel $\mathcal{T}$, reward operator R , and stationary state σ , put $q = \text{Tr}(\sigma R)$. On the centred subspace the Poisson equation

$$K - \mathcal{T}^*(K) = R - qI \quad (45)$$

has a solution under primitivity, giving

$$\sum_{t=0}^{N-1} \text{Tr}(\rho_t R) = Nq + \text{Tr}(\rho_0 K) - \text{Tr}(\rho_N K). \quad (46)$$

Thus the long-run witness is determined by q , while K is a bounded transient potential. If $R = qI$, the expected score is state-independent on every round and no reset is required.

D Exact contextual regeneration values

For the 13-ray state-independent witness, direct projector multiplication gives the scalar operator used in the protocol and an exact per-round separation of $1/75$ from the noncontextual limit. For the five-ray family, symmetry reduces the stationary state to one target-overlap coordinate. Selective replacement after a no result is optimal within the audited family. The thresholds are

$$r_{\text{no}}^* = 0.8396425434, \quad (47)$$

$$c_{\text{min}} = 0.5037855260, \quad (48)$$

$$r_{\text{uniform}}^* = 0.5115311845, \quad (49)$$

and yes-only replacement reaches only $1.9098300563 < 2$.

E Corrected four-record instrument

After expressing effects and returned states in one basis, the transition matrix is

$$C = \begin{pmatrix} 0.43696297 & 0.19810609 & 0.20644079 & 0.14060033 \\ 0.16017410 & 0.46165671 & 0.13951861 & 0.22362294 \\ 0.24443666 & 0.11892971 & 0.46061556 & 0.17659060 \\ 0.15842628 & 0.22130750 & 0.19342504 & 0.45918613 \end{pmatrix}. \quad (50)$$

Its stationary law is

$$\pi = (0.24327934, 0.24563854, 0.24991431, 0.26116781), \quad (51)$$

and every pair of columns has positive total-variation distance, proving that the four current records are the unique coarsest first-order predictive states. The exact information ledger is

$$H(A_{t+1}) = 1.9994586368, \quad (52)$$

$$H(A_{t+1} | A_t) = 1.8413629168, \quad (53)$$

$$I(A_t; A_{t+1}) = 0.1580957201 \quad (54)$$

bits. The finite-memory Landauer floor is therefore $1.2763355141 k_B T$ per use when the previous record is retained as side information.

Table 4: Decay of retained source information in the corrected repeated instrument.

reuse t	$I(J; A_t)$ bits	optimal source-guess accuracy
0	0.4720622220	0.6232838703
1	0.0409311197	0.3514199523
2	0.0040117838	0.2790063220
3	0.0004285525	0.2586660216
4	4.8954×10^{-5}	0.2526827310
10	1.7447×10^{-10}	0.2500040417

F Reproduction obligations

A full reproduction verifies projector identities, selected and unread negativity, transition probabilities, entropies, correlator decay, fixed spaces, Poisson telescoping, contextual bounds, threshold optimization, basis transformation, physical positivity repair, Kraus completeness, Choi positivity, stationary law, and information decay. Public matrix calculations are exact for the supplied processed estimates; population uncertainty requires the raw-count covariance.

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